

Design for class “Cart”

Cart
- items : List<OrderItem>
+ addItem(product : Product, quantity : int) : void + removeItem(product : Product) : void + getTotalWeight() : float + getSubtotal() : float + clearCart() : void

Table 1. Attribute Design

#	Name	Data type	Default value	Description
1	items	List	empty list	The collection of order items currently in the cart

Table 2. Operation Design

#	Name	Return type	Description / Purpose
1	addItem(product, quantity)	void	Add a product with specified quantity to the cart
2	removeItem(productId)	void	Remove an item from the cart using its product ID
3	updateQuantity(productId, qty)	void	Update the quantity of an item in the cart
4	getItems()	Collection	Return all items currently in the cart

Method Design:

1. addItem

- Parameter:
 - product: Product - the product to be added
 - quantity: int - the number of units to add

- Exception:

- Method:
 - Check if an OrderItem with the same product already exists.
 - If yes, increase its quantity by the input amount.
 - Otherwise, create a new OrderItem and add it to items.
 - No value is returned.

2. removeItem

- Parameter:
 - productId: String - the identifier of the product to remove

- Exception:

- Method:
 - Find the OrderItem whose product ID matches productId.
 - Remove that OrderItem from the items list.

3. getDeliveryInfo

- Parameter:
 - productId: String - the identifier of the product to update
 - qty: int - the new quantity

- Exception:
 - InvalidQuantityException if $qty \leq 0$

- Method:
 - Find the OrderItem whose product ID matches productId.
 - If qty is invalid (≤ 0), throw InvalidQuantityException.
 - Else update the OrderItem's quantity.

4. showInvalidInfoError

- Parameter:

- Exception:

- Method:

- Return the internal list of OrderItem objects.